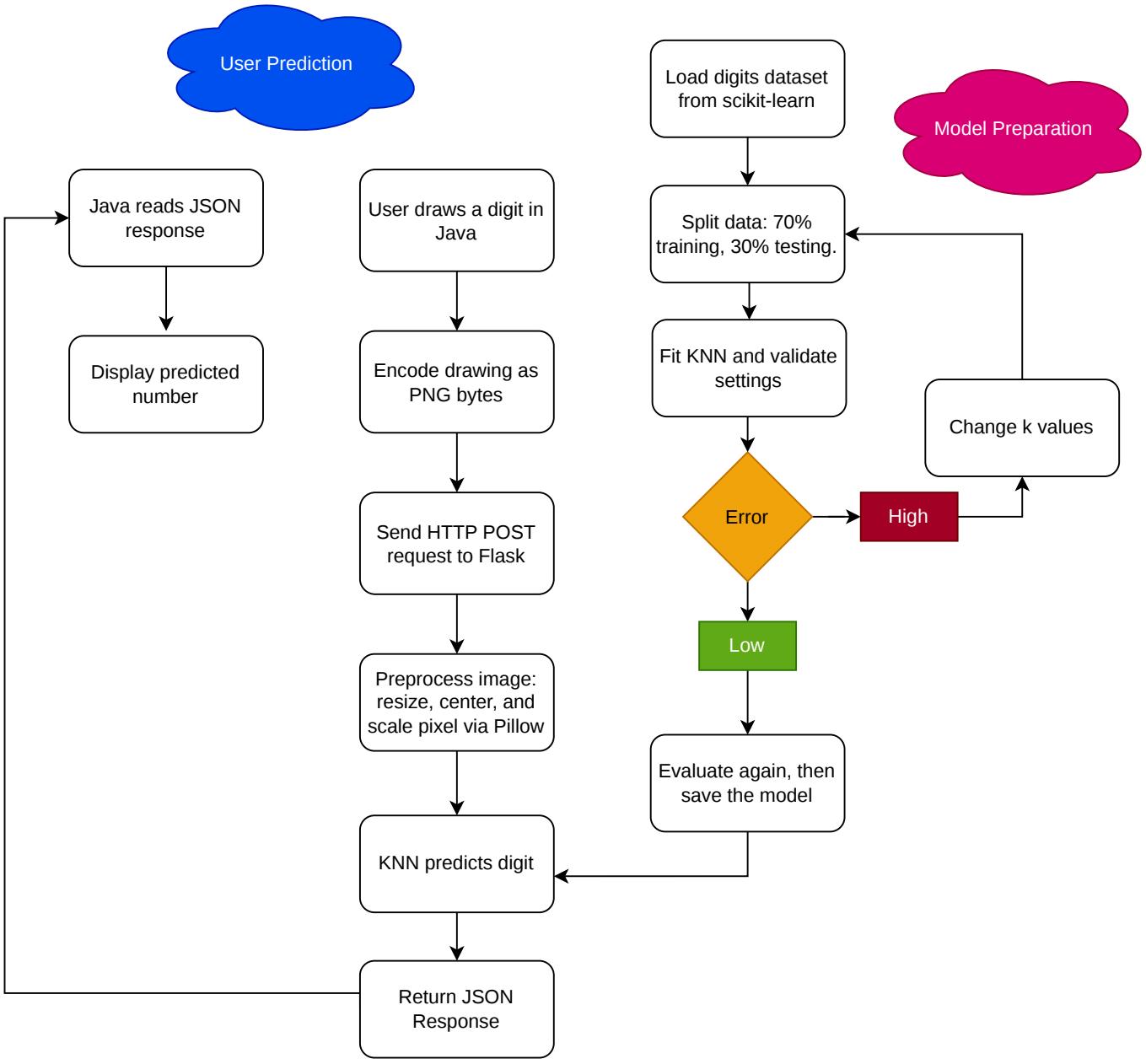
User Prediction
Java reads JSON response
Display predicted number
User draws a digit in Java
Encode drawing as PNG bytes
Send HTTP POST request to Flask
Preprocess image: resize, center, and scale pixel via Pillow
KNN predicts digit
Return JSON Response
Load digits dataset from scikit-learn
Model Preparation
Split data: 70% training, 30% testing.
Fit KNN and validate settings
Error
High
Change k values
Low
Evaluate again, then save the model